



MTE 4108

Vision Guided Parallel SCARA for Precise pick & place operation

Automated Sorting System

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Hardware Implementation

Core Components

Sensors

- Overhead Vision Camera
- Rotary Encoders and Magnetic Encoder

Actuation

Base-grounded NEMA 17 stepper motors with DRV8825 drivers (1/32 microstepping).

Mechanics

Custom 10 mm thick planar linkages for high Z-axis rigidity.

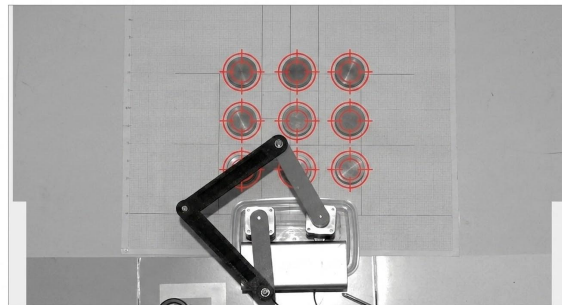
Control & Vision

ESP32 and ESP32-CAM.

End-Effector

5V Electromagnet with active-low relay isolation.

Physical SCARA Robot Setup



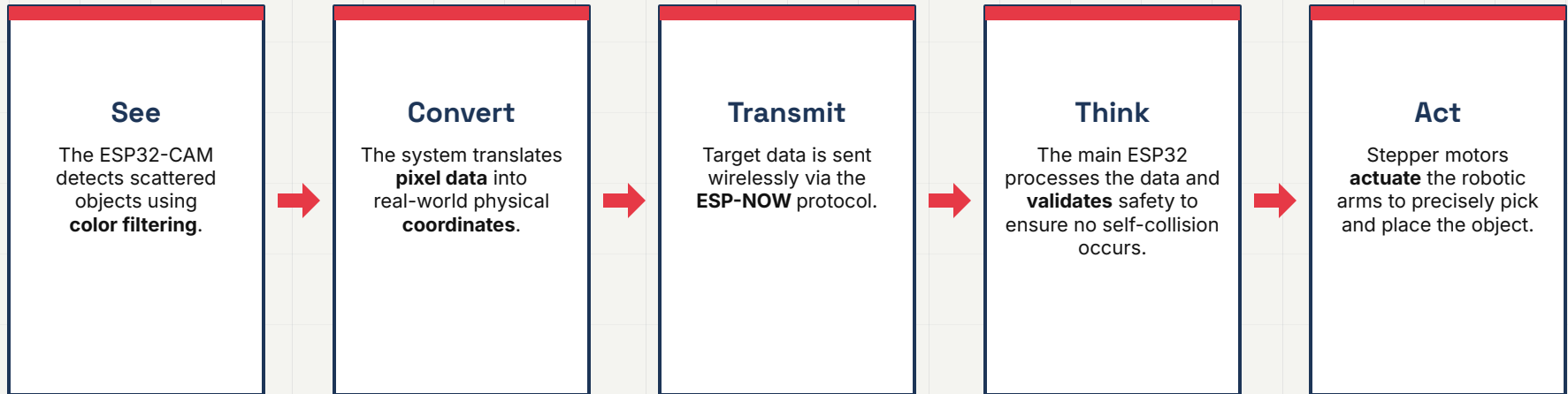
Prototype Integration

Real-world hardware implementation showing NEMA 17 actuators, planar links, and custom ESP32 control board.



Overall Execution Pipeline

The system operates through a highly efficient, five-step decentralized pipeline:



See (Object Detection)

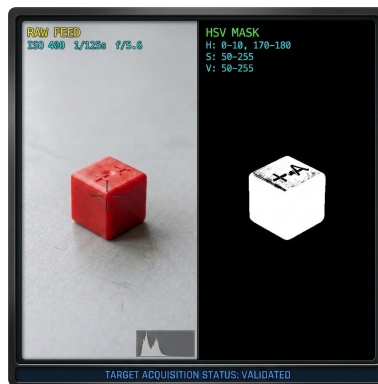
Workspace Capture & Segmentation

Overhead Capture: An overhead ESP32-CAM continuously monitors and captures the workspace environment in real-time.

Color Filtering: Lightweight, efficient color filtering algorithms detect scattered target objects dynamically.

HSV Isolation: Raw RGB data is converted to the HSV color space to completely isolate the target object from background noise entirely on the edge device.

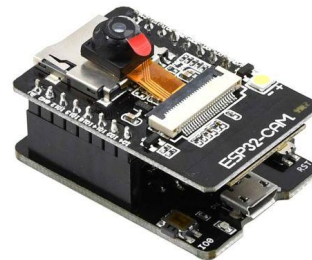
HSV Segmentation



Target Acquisition

Real-time feed alongside binary HSV mask isolating the target object.

ESP32-CAM Module



Edge Hardware

Compact, low-cost microcontroller performing on-device computer vision.

Convert – Spatial Mapping

Homography & Translation

Pixel-to-Physical Calibration

Centroid Detection

The vision node isolates the geometric center (centroid) of the detected object as a precise pixel value (U, V).

Mathematical Translation

The system mathematically translates this localized pixel data into real-world physical coordinates.

Planar Homography

A planar homography transformation matrix scales and translates optical data into physical Cartesian coordinates (X, Y) in millimeters.

Mapping Matrix Equation

$$\begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & C_x \\ 0 & S_y & C_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U \\ V \\ 1 \end{bmatrix}$$

Transformation Operator

Applies scaling factors (S) and translation offsets (C) to project digital optical sensor coordinates straight to physical workbench measurements.

Transmit – Wireless Telemetry

Direct Node-to-Node Link

Vision-to-Kinematics Link

The calculated target data is sent wirelessly from the vision node to the kinematic node in real-time.

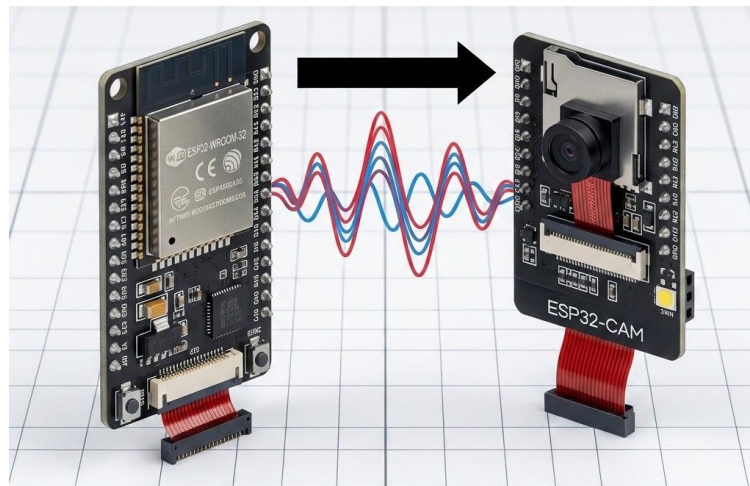
ESP-NOW Protocol

Data transmission relies on the connectionless **“ESP-NOW protocol”** for efficient, direct MAC-to-MAC communication.

Deterministic Pipeline

Bypassing traditional Wi-Fi routers guarantees a highly stable, deterministic, microsecond-latency data pipeline.

Wireless Telemetry Bridge



ESP32-CAM to Main ESP32

Low-latency, direct wireless link acting as the communication bridge between processing nodes.

Think – Kinematics & Safety



Kinematics & Safety Pipeline

1. Target Processing

The main ESP32 processes the received (X, Y) target data through the *inverse kinematics solver* in real-time.

2. Trajectory Validation

The controller mathematically validates the proposed trajectory to guarantee that no self-collision occurs during execution.

3. Linkage Verification

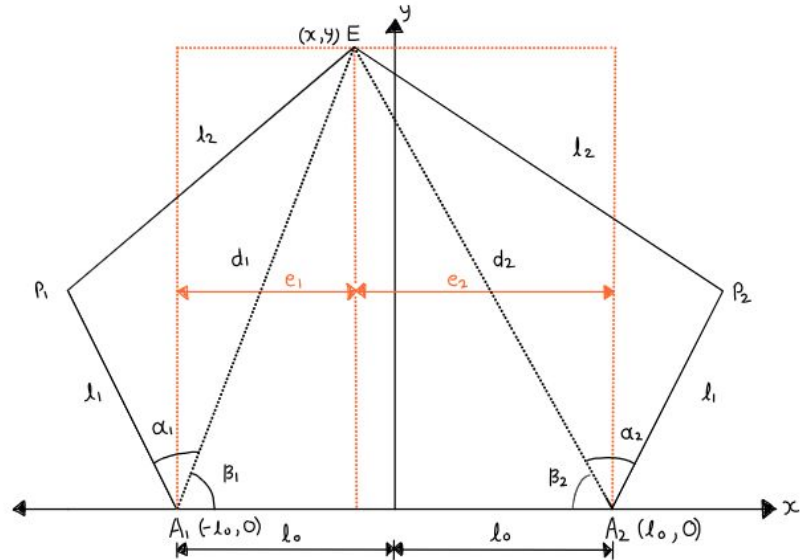
A 2-D cross-product line-intersection monitor verifies the safety of the 10 mm linkages before authorizing motor movement.

Mathematical Safety Constraint

$$v_{link1} \times v_{link2} = (x_1 y_2 - x_2 y_1) \neq 0$$

Real-time intersection monitoring prevents structural conflicts.

Kinematic Modelling (IK Solver)



Geometric parameters of 5 bar parallel SCARA Manipulator

Analytical Kinematic Equations

$$\alpha_1 = \cos^{-1} \left(\frac{l_1^2 + (l_0 + x)^2 + y^2 - l_2^2}{2l_1 \sqrt{(l_0 + x)^2 + y^2}} \right)$$

$$\alpha_2 = \cos^{-1} \left(\frac{l_1^2 + (l_0 - x)^2 + y^2 - l_2^2}{2l_1 \sqrt{(l_0 - x)^2 + y^2}} \right)$$

$$\beta_1 = \tan^{-1} \left(\frac{y}{l_0 + x} \right)$$

$$\beta_2 = \tan^{-1} \left(\frac{y}{l_0 - x} \right)$$

As **(+-) configuration** has been chosen for maximum reach and large workspace:

$$\theta_1 = \beta_1 + \alpha_1 \quad | \quad \theta_2 = \pi - \beta_2 - \alpha_2$$

Act – Physical Execution

Execution Pipeline

01

Driver Actuation

The validated step trajectories are pushed to the DRV8825 drivers to actuate the robotic arms via NEMA 17 stepper motors.

02

Velocity Profiling

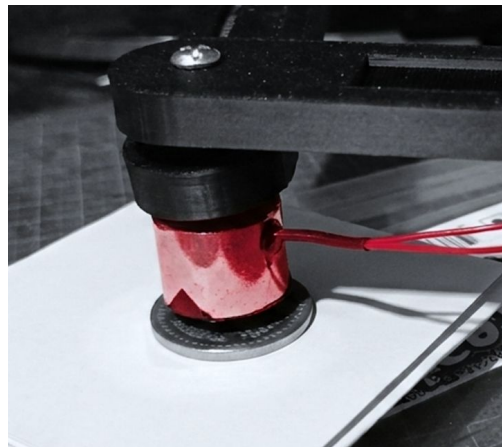
The system executes smooth, trapezoidal velocity profiles to precisely pick and place the target object.

03

Geometric Safety

The hardware maintains a strict $(-, +)$ elbow-out geometric configuration to navigate safely to the destination tray.

Physical Execution & End-Effector



Electromagnet Pick-and-Place

5V electromagnetic end-effector securely holds and places the target object dynamically.



Thank you